

```
SELECT
    InvoiceNo, StockCode, Quantity, UnitPrice,
    (Quantity*UnitPrice) AS amount
FROM
    `customer-segmentation-477415.retail.sales`
```

```
WITH bills AS
(
    SELECT
        InvoiceNo,
        (Quantity*UnitPrice) AS amount
    FROM
        `customer-segmentation.retail.sales`
)
SELECT
    InvoiceNo,
    SUM(amount) AS total
FROM bills
GROUP BY InvoiceNo
```

```
SELECT
    CustomerID,
    DATE(MAX(InvoiceDate)) AS last_purchase_date,
    DATE(MIN(InvoiceDate)) AS first_purchase_date,
    COUNT(DISTINCT InvoiceNo) AS num_purchases,
    SUM(total) AS monetary,
FROM (
    Select
        s.CustomerID, s.InvoiceDate, s.InvoiceNo, b.total,
        ROW_NUMBER() OVER(PARTITION BY s.InvoiceNo ORDER BY s.InvoiceNo) AS RN
    FROM `customer-segmentation-477415.retail.sales` s
    LEFT JOIN `customer-segmentation-477415.retail.bills` b
        ON s.InvoiceNo=b.InvoiceNo
    ) A
WHERE A.RN = 1
GROUP BY CustomerID
```

```
SELECT *,
    DATE_DIFF(reference_date, last_purchase_date, DAY) AS recency,
    num_purchases/ (months_cust) AS frequency,
FROM (
    SELECT *,
        MAX(last_purchase_date) OVER () + 1 AS reference_date,
        DATE_DIFF(last_purchase_date, first_purchase_date, month)+1 AS months_cust
    FROM `customer-segmentation-477415.retail.monetary`
)
ORDER BY CustomerID
```

```
SELECT
    a.*,
    --All percentiles for MONETARY
    b.percentiles[offset(20)] AS m20,
    b.percentiles[offset(40)] AS m40,
    b.percentiles[offset(60)] AS m60,
    b.percentiles[offset(80)] AS m80,
    b.percentiles[offset(100)] AS m100,
    --All percentiles for FREQUENCY
    c.percentiles[offset(20)] AS f20,
    c.percentiles[offset(40)] AS f40,
    c.percentiles[offset(60)] AS f60,
    c.percentiles[offset(80)] AS f80,
    c.percentiles[offset(100)] AS f100,
    --All percentiles for RECENCY
    d.percentiles[offset(20)] AS r20,
    d.percentiles[offset(40)] AS r40,
    d.percentiles[offset(60)] AS r60,
    d.percentiles[offset(80)] AS r80,
    d.percentiles[offset(100)] AS r100

FROM
```

```

`retail.RFM` a,

(

SELECT APPROX_QUANTILES(monetary, 100) percentiles
FROM `retail.RFM`) b,
(SELECT APPROX_QUANTILES(frequency, 100) percentiles
FROM `retail.RFM`) c,
(SELECT APPROX_QUANTILES(recency, 100) percentiles
FROM `retail.RFM`
)d

ORDER BY CustomerID

```

```

SELECT
  CustomerID,
  m_score, f_score, r_score,
  recency, frequency, monetary,
  CAST(ROUND((f_score + m_score) / 2, 0) AS INT64) AS fm_score
FROM (
  SELECT *,
  CASE WHEN monetary <= m20 THEN 1
        WHEN monetary <= m40 AND monetary > m20 THEN 2
        WHEN monetary <= m60 AND monetary > m40 THEN 3
        WHEN monetary <= m80 AND monetary > m60 THEN 4
        WHEN monetary <= m100 AND monetary > m80 THEN 5
  END AS m_score,
  CASE WHEN frequency <= f20 THEN 1
        WHEN frequency <= f40 AND frequency > f20 THEN 2
        WHEN frequency <= f60 AND frequency > f40 THEN 3
        WHEN frequency <= f80 AND frequency > f60 THEN 4
        WHEN frequency <= f100 AND frequency > f80 THEN 5
  END AS f_score,
  --Recency scoring is reversed
  CASE WHEN recency <= r20 THEN 5
        WHEN recency <= r40 AND recency > r20 THEN 4
        WHEN recency <= r60 AND recency > r40 THEN 3
        WHEN recency <= r80 AND recency > r60 THEN 2
        WHEN recency <= r100 AND recency > r80 THEN 1
  END AS r_score,
  FROM `retail.quantile`
)

```

```

SELECT CustomerID, count(distinct InvoiceNo) FROM `customer-segmentation-477415.retail.sales`

group by CustomerID;

```

```

SELECT
  InvoiceNo, StockCode, Quantity, UnitPrice,
  (Quantity*UnitPrice) AS amount
FROM
`customer-segmentation-477415.retail.sales`

```

```

WITH bills AS
(
  SELECT
    InvoiceNo,
    (Quantity*UnitPrice) AS amount
  FROM
    `customer-segmentation-477415.retail.sales`
)
SELECT
  InvoiceNo,
  SUM(amount) AS total
FROM bills
GROUP BY InvoiceNo

```

```

SELECT
  CustomerID,
  DATE(MAX(InvoiceDate)) AS last_purchase_date

```

```

DATE(MAX(InvoiceDate)) AS last_purchase_date,
DATE(MIN(InvoiceDate)) AS first_purchase_date,
COUNT(DISTINCT InvoiceNo) AS num_purchases,
SUM(total) AS monetary,
FROM (
    Select
        s.CustomerID, s.InvoiceDate, s.InvoiceNo, b.total,
        ROW_NUMBER() OVER(PARTITION BY s.InvoiceNo ORDER BY s.InvoiceNo) AS RN
    FROM `customer-segmentation-477415.retail.sales` s
    LEFT JOIN `customer-segmentation-477415.retail.bills` b
    ON s.InvoiceNo=b.InvoiceNo
) A
WHERE A.RN = 1
GROUP BY CustomerID

SELECT *,
    DATE_DIFF(reference_date, last_purchase_date, DAY) AS recency,
    num_purchases/ (months_cust) AS frequency,
FROM (
    SELECT *,
        MAX(last_purchase_date) OVER () + 1 AS reference_date,
        DATE_DIFF(last_purchase_date, first_purchase_date, month)+1 AS months_cust
    FROM `customer-segmentation-477415.retail.monetary`
)
ORDER BY CustomerID

SELECT
    a.*,
    --All percentiles for MONETARY
    b.percentiles[offset(20)] AS m20,
    b.percentiles[offset(40)] AS m40,
    b.percentiles[offset(60)] AS m60,
    b.percentiles[offset(80)] AS m80,
    b.percentiles[offset(100)] AS m100,
    --All percentiles for FREQUENCY
    c.percentiles[offset(20)] AS f20,
    c.percentiles[offset(40)] AS f40,
    c.percentiles[offset(60)] AS f60,
    c.percentiles[offset(80)] AS f80,
    c.percentiles[offset(100)] AS f100,
    --All percentiles for RECENCY
    d.percentiles[offset(20)] AS r20,
    d.percentiles[offset(40)] AS r40,
    d.percentiles[offset(60)] AS r60,
    d.percentiles[offset(80)] AS r80,
    d.percentiles[offset(100)] AS r100

FROM
    `retail.RFM` a,

(

SELECT APPROX_QUANTILES(monetary, 100) percentiles
    FROM `retail.RFM`) b,
(SELECT APPROX_QUANTILES(frequency, 100) percentiles
    FROM `retail.RFM`) c,
(SELECT APPROX_QUANTILES(recency, 100) percentiles
    FROM `retail.RFM`
)d

ORDER BY CustomerID

SELECT
    CustomerID,
    m_score, f_score, r_score,
    recency, frequency, monetary,
    CAST(ROUND((f_score + m_score) / 2, 0) AS INT64) AS fm_score
FROM (
    SELECT *,
        CASE WHEN monetary <= m20 THEN 1
            WHEN monetary <= m40 AND monetary > m20 THEN 2
            WHEN monetary <= m60 AND monetary > m40 THEN 3
            WHEN monetary <= m80 AND monetary > m60 THEN 4

```

```

    WHEN monetary <= m100 AND monetary > m80 THEN 5
END AS m_score,
CASE WHEN frequency <= f20 THEN 1
    WHEN frequency <= f40 AND frequency > f20 THEN 2
    WHEN frequency <= f60 AND frequency > f40 THEN 3
    WHEN frequency <= f80 AND frequency > f60 THEN 4
    WHEN frequency <= f100 AND frequency > f80 THEN 5
END AS f_score,
--Recency scoring is reversed
CASE WHEN recency <= r20 THEN 5
    WHEN recency <= r40 AND recency > r20 THEN 4
    WHEN recency <= r60 AND recency > r40 THEN 3
    WHEN recency <= r80 AND recency > r60 THEN 2
    WHEN recency <= r100 AND recency > r80 THEN 1
END AS r_score,
FROM `retail.quantile`
)

SELECT
CustomerID,
--recency, frequency, monetary,
--r_score, f_score, m_score,
--fm_score,
CASE
    WHEN (r_score = 5 AND fm_score = 5)
        OR (r_score = 5 AND fm_score = 4)
        OR (r_score = 4 AND fm_score = 5)
    THEN 'Champions'
    WHEN (r_score = 5 AND fm_score = 3)
        OR (r_score = 4 AND fm_score = 4)
        OR (r_score = 3 AND fm_score = 5)
        OR (r_score = 3 AND fm_score = 4)
    THEN 'Loyal Customers'
    WHEN (r_score = 5 AND fm_score = 2)
        OR (r_score = 4 AND fm_score = 2)
        OR (r_score = 3 AND fm_score = 3)
        OR (r_score = 4 AND fm_score = 3)
    THEN 'Potential Loyalists'
    WHEN r_score = 5 AND fm_score = 1
    THEN 'Recent Customers'
    WHEN (r_score = 4 AND fm_score = 1)
        OR (r_score = 3 AND fm_score = 1)
    THEN 'Promising'
    WHEN (r_score = 3 AND fm_score = 2)
        OR (r_score = 2 AND fm_score = 3)
        OR (r_score = 2 AND fm_score = 2)
    THEN 'Customers Needing Attention'
    WHEN r_score = 2 AND fm_score = 1
    THEN 'About to Sleep'
    WHEN (r_score = 2 AND fm_score = 5)
        OR (r_score = 2 AND fm_score = 4)
        OR (r_score = 1 AND fm_score = 3)
    THEN 'At Risk'
    WHEN (r_score = 1 AND fm_score = 5)
        OR (r_score = 1 AND fm_score = 4)
    THEN 'Cant Lose Them'
    WHEN r_score = 1 AND fm_score = 2 THEN 'Hibernating'
    WHEN r_score = 1 AND fm_score = 1 THEN 'Lost'
END AS rfm_segment
FROM `retail.score`
ORDER BY CustomerID

```

